

Member ID: _____

Time: _____

Rank: _____



DEVICE CONFIGURATION AND TROUBLESHOOTING (305) **STATE 2023**

Total Points (50 @ 2 points each) _____ **(100 points)**

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. A user purchased a stick of RAM and want to know if it is compatible with their system. The label says “PC3-12800/8GB.” They have a system board that requires DDR4 memory. Is this memory compatible with this system?
 - A. Yes
 - B. No
 - C. Only if they are running a 64-Bit Operating System
 - D. Only if the memory uses ECC
2. Windows 10 requires how much system memory for a 64-Bit Installation (minimum)?
 - A. 8GB
 - B. 2GB
 - C. 4GB
 - D. 1GB
3. A user is wanting to install Windows 11 on their system. What is not a requirement for Windows 11 install
 - A. 1 GHz processor
 - B. TPM module version 2.0
 - C. 1 TB of HDD space
 - D. Secure boot capable UEFI BIOS
4. RAID 5 arrays require a minimum of how many disks?
 - A. 3 disks
 - B. 2 disks
 - C. 5 disks
 - D. 4 disks
5. True or False: When a permission is assigned to a folder, that permission gets applied also to its subfolders.
 - A. True
 - B. False
6. This measurement unit is used to measure the speed of system components and represents the number of “clock” cycles per second they perform.
 - A. BPS
 - B. Volts
 - C. Hertz
 - D. RPS

7. Which of the following statements best describes a Simple Network Switch?
 - A. Redirects traffic from one network to another
 - B. Repeats all traffic sent to one port out of all other ports
 - C. Directs traffic by examining its MAC address and sending to the port where that addresses owner is located
 - D. Directs traffic by examining its IP address and sending to the port where that addresses owner is located
8. Convert the number 99 to binary
 - A. 1011110
 - B. 1111110
 - C. 1100011
 - D. 0010010
9. What tool allows a user to see diagnostic data about a system before it boots into an Operating System?
 - A. Hardware Investigator (H-INV)
 - B. UEFI
 - C. Windows Startup Recovery
 - D. POST card
10. What type of RAID setup is commonly used for fault tolerance?
 - A. RAID 0
 - B. RAID 1
 - C. RAID 5
 - D. RAID 6
11. What Linux command will change the directory you're viewing?
 - A. mkdir
 - B. cd
 - C. chmod
 - D. ls
12. Which of the following is a Class A subnet mask?
 - A. 255.255.0.0
 - B. 255.0.0.0
 - C. 255.255.255.0
 - D. 255.255.254.0

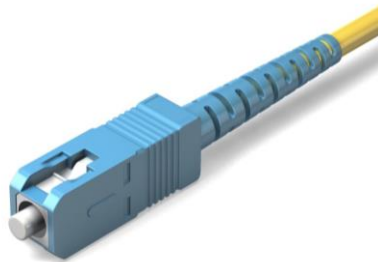
13. After power off, what is the first thing you should do before opening a portable computer for repairs?
 - A. Connect an anti-static strap to bare metal
 - B. Remove the battery
 - C. Hold the power button down for 30 seconds to drain power
 - D. Place parts on an anti-static mat
14. You keep commonly failed component shelf spare parts in storage for quick and easy repair of the systems you administrate. What is the best way to store these components in order to protect them? (Choose the best answer)
 - A. On top of an anti-electrostatic bag
 - B. In a non-air conditioned warehouse
 - C. Inside of an anti-electrostatic bag
 - D. Inside an anti-electrostatic bag in a climate controlled area
15. In a modern day Linux distribution, what is the most default filesystem?
 - A. Ext3
 - B. Ext4
 - C. Ext64
 - D. Ext32
16. What Linux command is used to change permissions on a file or folder
 - A. chmod
 - B. mkdir
 - C. ls
 - D. Sudo
17. If you need to run a command in Linux as admin, what is the first thing you should type in your command
 - A. su
 - B. chmod
 - C. sudo
 - D. cat
18. You have been given a system that does not boot stating that no boot device is available. What is an easy first step to troubleshoot?
 - A. Replace the system drive
 - B. Change the boot mode from BIOS to UEFI
 - C. Reformat the drive and reinstall the OS
 - D. Replace the system power cord

19. Which of the following acts as a sort of “address book” that translates web addresses into IP addresses?
 - A. DNS
 - B. DHCP
 - C. TCP/IP
 - D. MAC Address Table
20. How should an anti-static wrist bracelet be used when working on components inside a computer?
 - A. Connect the bracelet to your rubber-soled shoes
 - B. Connect the bracelet to the computer case
 - C. Connect the bracelet to a capacitor in your pocket
 - D. Connect the bracelet to anything metal nearby
21. What type of printer uses negatively charged areas on a metallic drum to pick up toner which is then bonded to the paper using pressure from a heated roller?
 - A. impact
 - B. inkjet
 - C. laser
 - D. thermal
22. Which partition type supports drives larger than 2 TB?
 - A. GPT
 - B. FAT32
 - C. MBR
 - D. None of these
23. What Windows command would you use to locate and consolidate fragmented files to increase system performance?
 - A. sfc
 - B. chkdsk
 - C. defrag
 - D. diskpart
24. On your computer, you have a C drive for Windows files and a D drive for other data. After a drive failure on the system, you discover you lost all data on both the C and D drives. What is the most likely reason for this?
 - A. The drives were part of a RAID-0
 - B. Both drives were partitions on the same disk
 - C. Both drives failed at the same time
 - D. Radiation exposure corrupted the data

25. You have obtained a new external hard drive that will migrate between both Windows 10 and Ubuntu systems. What file system should be selected to format the new drive?
- A. APFS
 - B. Ext4
 - C. HFS+
 - D. NTFS
26. Which protocol is most associated with the Internet?
- A. NETBIOS
 - B. POP3
 - C. FTP
 - D. Telnet
27. Which of the following technologies allows new hardware to run older applications and operating systems that might otherwise be incompatible?
- A. Compatibility Mode
 - B. DOS Mode
 - C. Dual Boot
 - D. Virtual Machines
28. After an installation of Windows is completed, a technician opens Device Manager and finds several components listed with yellow triangles with exclamation marks. Based on this, what step might they need to complete before using this system further?
- A. Install the system drivers
 - B. Defragment the hard drive
 - C. Install another version of Windows
 - D. Install a virus scanner
29. Convert the number 99 to binary
- A. 1011110
 - B. 1111110
 - C. 1100011
 - D. 0010010
30. You have a client system that has the error “The trust relationship between this workstation and the primary domain failed.” How should you resolve this issue
- A. Reinstall the OS on the machine
 - B. Disjoin and rejoin the domain
 - C. Reset the user’s password in Active Directory Users and Computers
 - D. Reboot the system

31. Processors using a grid of flat metal contacts to connect to blunt pins on the system board are using a socket type called:
- A. Flat Grid Array
 - B. Contact Grid Array
 - C. Land Grid Array
 - D. Zero Insertion Force Array
32. System memory capable of recovering from a single-bit error is using which of the following technologies?
- A. Internal Parity Correction
 - B. Automatic Registry Cleaning
 - C. Memory Corruption Detection
 - D. Error Correction Code

33. What is this a picture of:



- A. Coaxial Cable
 - B. RJ-45
 - C. Fiber Optic Cable
 - D. Component cable
34. When troubleshooting a system that will not complete POST or display video, which of the following steps should you perform first before attempting to power on again?
- A. Remove all system memory
 - B. Remove the internal SATA hard drive
 - C. Remove the USB Keyboard
 - D. Remove the video card
35. True or false: A system needs a mouse and keyboard to complete POST and boot
- A. True
 - B. False
36. When no DHCP server is available, which protocol assigns an IP address to network ports in Windows when no static IP address is set?
- A. APIPA
 - B. DNS
 - C. NETBIOS
 - D. This is not possible without DHCP

37. Which of the following is **not** a valid file system available within Windows 10:
- A. FAT
 - B. FAT32
 - C. FAT64
 - D. NTFS
38. Which of the following reasons would you most likely use the **tracert** command?
- A. Quickly test connectivity between devices
 - B. List IP addresses and port numbers
 - C. Enable packet tracing on an interface
 - D. Determine how many hops are between devices
39. Which of the following reasons would you most likely use the **diskpart** command?
- A. Scan disk partitions for errors
 - B. List disks, partitions, and volumes
 - C. Modify the boot configuration on the disk
 - D. Enable encryption on the disk
40. Which of the following Linux and PowerShell commands could be considered an alias for the `dir` or `Get-ChildItem` command?
- A. `ls`
 - B. `cd`
 - C. `set`
 - D. `psdir`
41. Which of the following is not a good or valid reason to format a drive through Disk Management?
- A. Restore data that was previously overwritten
 - B. Extend an existing partition
 - C. Mark the disk as active
 - D. All of the above
42. You are working as an IT administrator for a small company. You have been asked to apply Windows Updates and after successfully updating five Windows 10 desktops, there is a Windows 10 desktop that is failing to reach Windows. Which of the following could be a valid troubleshooting step?
- A. Boot to Last Known Good Configuration
 - B. Boot to Safe Mode
 - C. Attempt to perform an automatic repair
 - D. All of the above
43. Which of the following is a feature or function of the `chkdsk` command?
- A. Scans the integrity of all protected system files and attempts to repair
 - B. Locates bad sectors and recovers readable information
 - C. Locates and consolidates fragmented files for performance improvement
 - D. Removes all partitions or volumes

44. Which of the following subnet masks have been subnetted?
- A. 255.255.255.255
 - B. 255.0.0.0
 - C. 255.255.254.0
 - D. None of the above
45. Which of the following is **not** required for communication or connectivity between two Windows devices that are connected through a switch?
- A. IP address
 - B. Subnet Mask
 - C. Default Gateway
 - D. B and C are not required
46. Which of the following command syntax would be valid to run System File Checker during a no boot situation or within the Windows Recovery Environment?
- A. SFC /scannow /offbootdir=C:\ /offwindir=C:\Windows
 - B. SFC /scannow
 - C. SFC /scannow /offbootdir=X:\ /offwindir=X:\Windows
 - D. SFC /scanfolder=C:\Windows
47. Which of the following command syntax would be valid to run DISM to attempt to revert pending updates during a no boot situation or within the Windows Recovery Environment?
- A. DISM /Online /Cleanup-Image /RevertPendingActions
 - B. DISM /Online /Cleanup-Image /RestoreHealth
 - C. DISM /Image:C:\ /Cleanup-Image /RevertPendingActions
 - D. DISM /Image:C:\ /Cleanup-Image /RestoreHealth
48. Your personal computer is hosting a server for a video game. The process used to start the server is supposed to listen on port 12345, but the video game has not been able to connect to the server over that port number. Which of the following commands could be used to see if the process is listening on port 12345?
- A. Nslookup
 - B. Netstat -ano
 - C. Ipconfig /all
 - D. Net use
49. You are working as an IT administrator for a small company. There is a user that manually connects to a shared folder on the file server each morning. Which of the following could be done to associate the shared folder with a drive letter, so that the user automatically has access?
- A. Create a mapped drive for [\\sharename\servername](#)
 - B. Create a mapped drive for [\\servername\sharename](#)
 - C. Create a mapped drive for [//sharename/servername](#)
 - D. Create a mapped drive for [//servername/sharename](#)

50. Which of the following drive types is the fastest?

- A. SAS
- B. NVMe
- C. SSD
- D. SATA